TEACH YOURSELF INTERMEDIATE XII UNIT 2

Computational Thinking & Algorithms (Advanced)

Tracing, efficiency, and correctness

What you will master

- Trace algorithms with trace tables
- Evaluate clarity and efficiency of solutions
- Compare algorithm strategies for the same problem

How to use these notes

- Read one section, then close the book and explain it aloud.
- Do every *Try it* box before moving on.
- Memorize tables and truth patterns — exams love them.

2.1 Correctness & tracing

A trace table records variables row-by-row for each statement — proves an algorithm works or reveals bugs.

```
// swap max of two
if a < b:
    temp = a; a = b; b = temp
```

Columns: line number, inputs, each variable, output.

2.2 Efficiency

Compare growth: constant , linear , quadratic , logarithmic .

Choose binary search over linear when data is sorted and large.

2.3 Refinement strategies

Start brute force → identify repeated work → add sorting/index → consider edge cases (empty list, duplicates).

Teach Yourself — Practice & Check

1. Trace selection sort on four elements with a table.
2. Why is risky for ?